

Building intelligent Global Health and
Humanitarian Response Technologies

Week 4:

Building at the Speed of Infection

Context Ebola: Engineering Trust

Engineering Listening to the World: *Epidemic Intelligence* at Scale

LiGHT

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EPFL

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**WORLD
ECONOMIC
FORUM**

Rwanda
Centre for the
Fourth Industrial
Revolution

How do you *detect* a catastrophe before it becomes one...and convince others to act on it...equitably?

- The fastest signal is not always the truest one.
- A dashboard can detect a pattern; only people and institutions can turn it into care.
- Detection is never neutral: every system decides what counts as a signal, whose data count, and whose absence is ignored.

Ebola: Engineering *Trust*

Lecturer: *Prof. Annie Hartley*

Light

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Outline

- 01 The outbreak before the outbreak
- 02 How surveillance sees: cases, signals, and rumours
- 03 Ebola: a map, a contact list, and a genome
- 04 Measuring spread under uncertainty
- 05 AI as early warning: promise, bias, and accountability

Ebola discovery

1976: Patient zero?

A nun falls ill near the **Ebola river** in Zaire (DRC) with an unexplained illness

Testing begins...

Her blood sample is sent to Antwerp in a blue thermos flask



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Ebola discovery

Everything is negative

- Lassa fever
- Yellow fever
- Malaria
- Marburg??

Electron microscopy

Threadlike: “filovirus”



but...Antigenic testing for Marberg = negative

Antigenically distinct...



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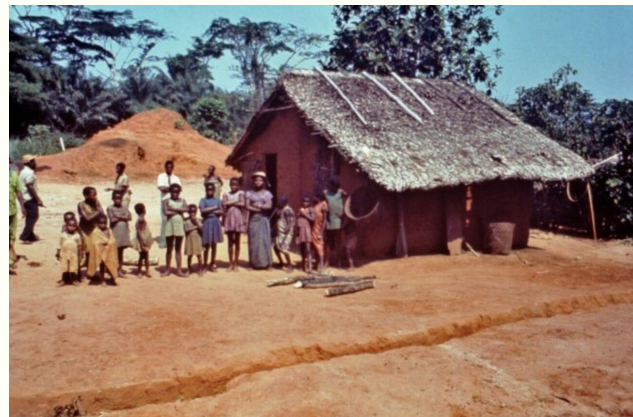
Groen (zittend voor microscoop), dr. Peter Piot en prof. dr. Pattyn
-koorts waarschijnlijk uit Soedan naar Zaïre overgebracht... (H)

Ebola discovery

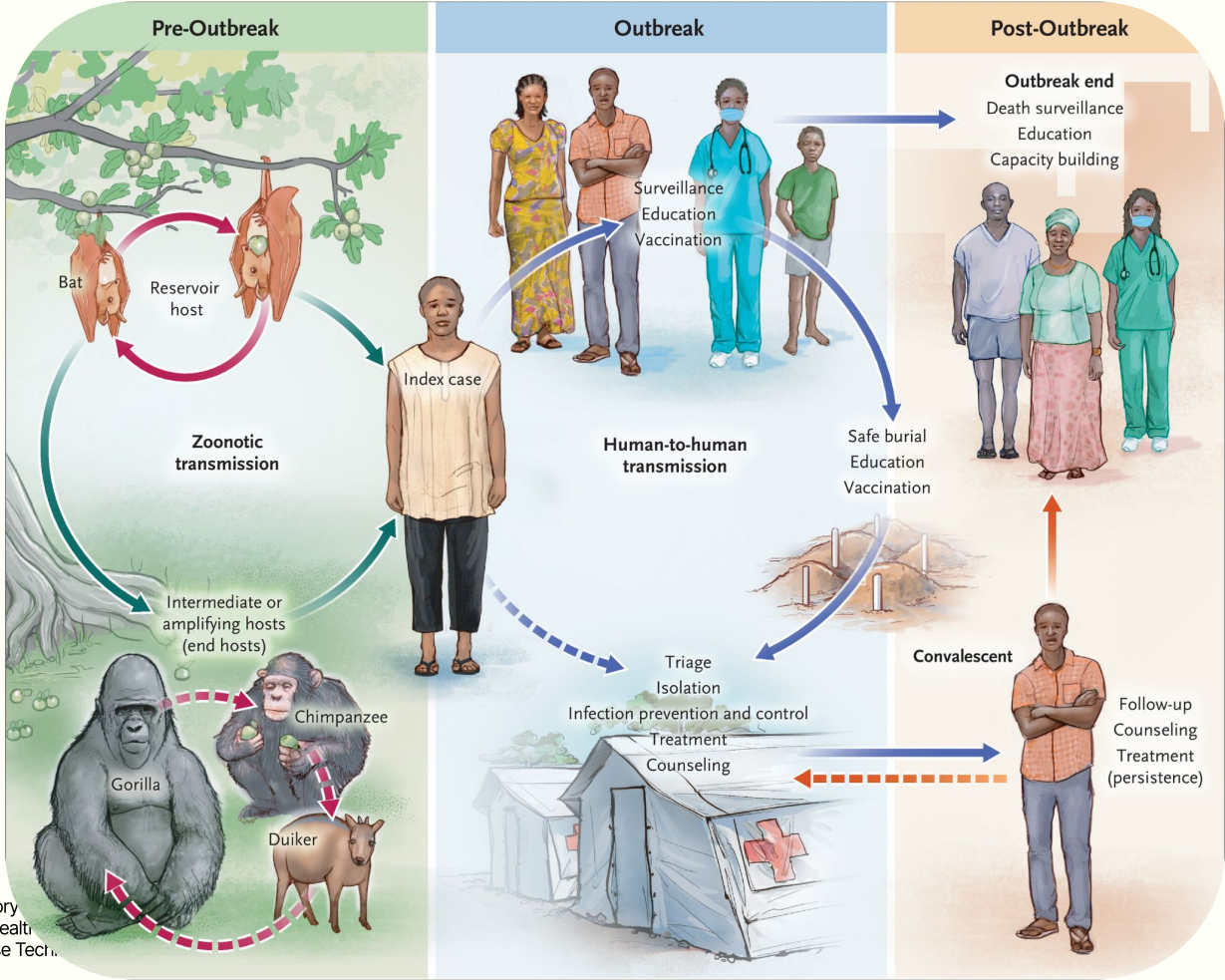
Search for patients

Transmission history

Patient characteristics



Transmission of Ebola

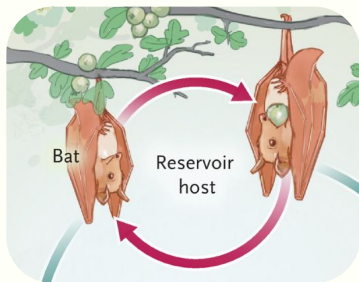


CONTEXT

Transmission of Ebola

Reservoir

Maintains the infection in nature over time



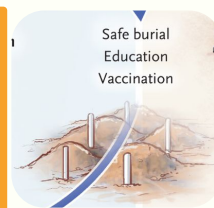
Zoonosis

An infection that moves from animals into people.



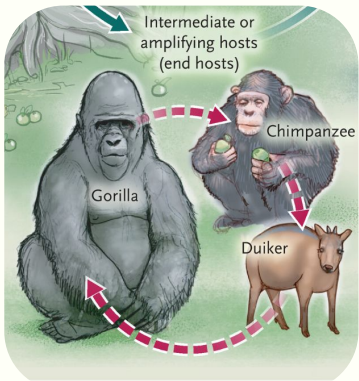
Fomite

An inanimate object that can transmit infection:
e.g. a doorknobs, bedsheet, surgical instrument, etc.



Intermediate hosts

An affected transmitter (is infected and may die)



Spillover

The moment a pathogen crosses from its usual animal host into a new host population

Vector

Unknown for Ebola

A living transmitter that does NOT get ill—usually an arthropod.
e.g. Mosquitoes: malaria, dengue, and yellow fever.

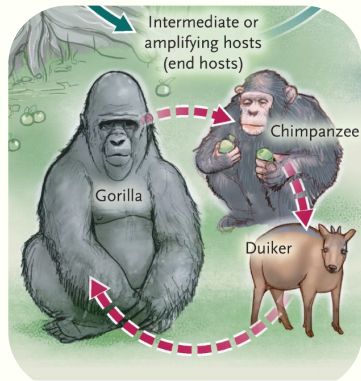
Transmission of Ebola

- Ebola has killed thousands of wild gorillas.
- Outbreaks beginning in 2002 and 2003 killed an estimated 5,000 gorillas in and around Lossi Sanctuary in the Republic of the Congo.

Bermejo et al. (2006). "Ebola Outbreak Killed 5000 Gorillas." Science, 314, 1564. doi:10.1126/science.1133105.

Intermediate hosts

An affected transmitter (is infected and may die)



Transmission of Ebola



The burial boys of Ebola https://www.youtube.com/watch?v=_3o7EJcN4Rk

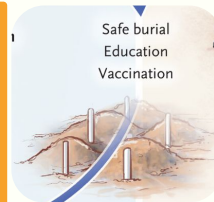
An Ebola Burial in DRC: <https://www.nytimes.com/2026/07/01/world/africa/ebola-congo-burials.html>

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Fomite

An inanimate object that can transmit infection:
e.g. a doorknobs, bedsheet, surgical instrument, etc.



Typhoid mary



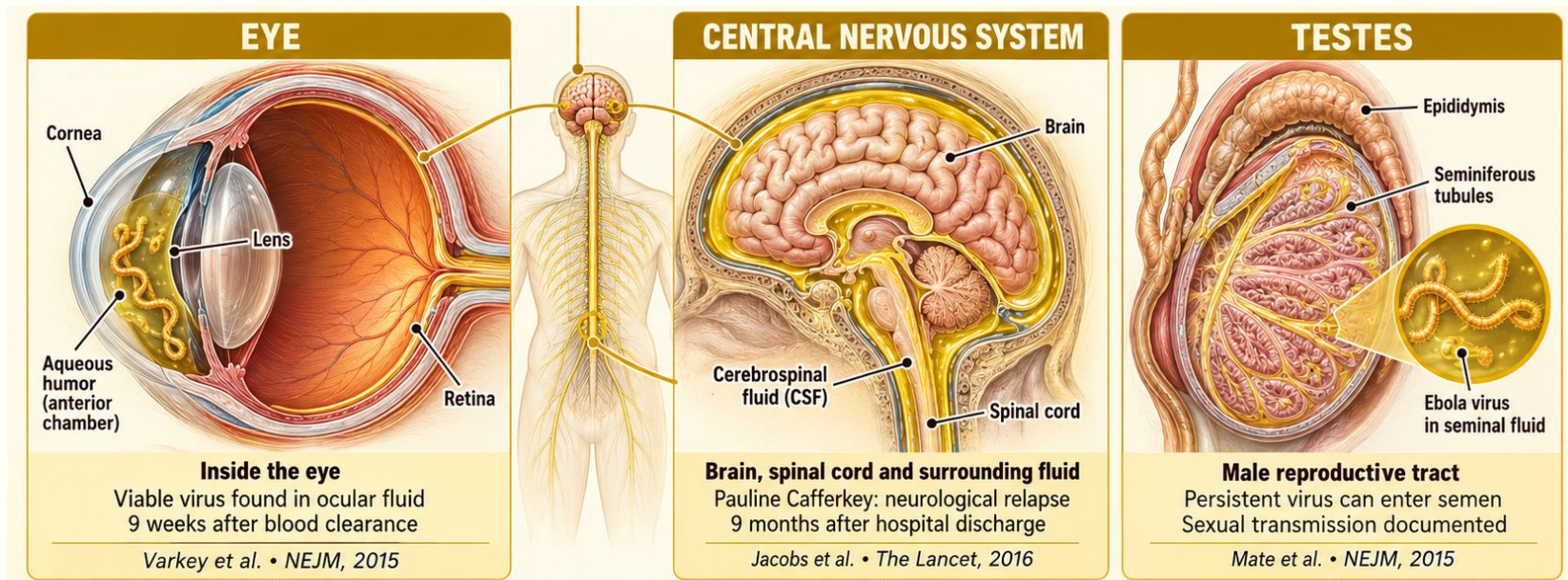
- Chronic carrier of *Salmonella Typhi*.
- **1907:** New York authorities confine her on North Brother Island
- **1910:** Released on condition that she not cook
→ Changes her name and works as a cook at Sloane Maternity Hospital
→ Outbreak follows
- **1915:** Reconfined on an island
→ until her death in 1938

Terminology

Carrier

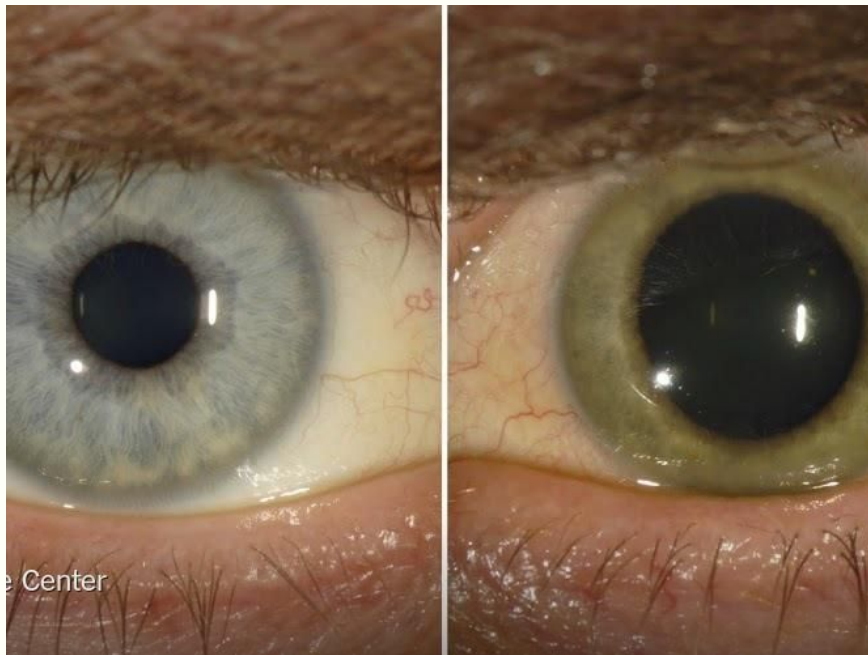
A reservoir in a Human population

Ebola carriers?



Transmission

Ebola carriers?



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The NEW ENGLAND
JOURNAL of MEDICINE

CURRENT ISSUE ▾ SPECIALTIES ▾ TOPICS ▾

ORIGINAL ARTICLE | BRIEF REPORT



Persistence of Ebola Virus in Ocular Fluid during Convalescence

 This article has been corrected. [VIEW THE CORRECTION](#)

Authors: Jay B. Varkey, M.D., Jessica G. Shantha, M.D., Ian Crozier, M.D., Colleen S. Kraft, M.D., G. Marshall Lyon, M.D., Aneesh K. Mehta, M.D., Gokul Kumar, M.D., , and Steven Yeh, M.D. [Author Info & Affiliations](#)

Published June 18, 2015 | N Engl J Med 2015;372:2423-2427 | DOI: 10.1056/NEJMoa1500306 | [VOL. 372 NO. 25](#)

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Transmission

Ebola carriers?

THE LANCET
Infectious Diseases

Multidisciplinary assessment of post-Ebola sequelae in Guinea (Postebogui): an observational cohort study

[Jean-François Etard, PhD](#) ^{a,*} · [Mamadou Saliou Sow, MD](#) ^{b,*} · [Sandrine Leroy, PhD](#) ^{a,c,*} · [Abdoulaye Touré, PhD](#) ^{a,d,*} · [Bernard Taverne, PhD](#) ^a · [Alpha Kabinet Keita, PhD](#) ^a · et al. [Show more](#)

Findings

Between March 23, 2015, and July 11, 2016, we recruited 802 patients, of whom 360 (45%) were male, 442 (55%) were female; 158 (20%) were younger than 18 years. The median age was 28·4 years (range 1·0–79·9, IQR 19·4–39·8). The median delay after discharge was 350 days (IQR 223–491). The most frequent symptoms were general symptoms (324 [40%] patients), musculoskeletal pain (303 [38%]), headache (278 [35%]), depression (124 [17%] of 713 responses), abdominal pain (178 [22%]), and [ocular disorders](#) (142 [18%]). More adults than children had at least one clinical symptom (505 [78%] vs 101 [64%], $p<0\cdot0003$), ocular complications (124 [19%] vs 18 [11%], $p=0\cdot0200$), or musculoskeletal symptoms (274 [43%] vs 29 [18%], $p<0\cdot0001$). A positive RT-PCR in semen was found in ten (5%) of 188 men, at a maximum of 548 days after disease onset. 204 (26%) of 793 patients reported stigmatisation. Ocular complications were more frequent at enrolment than at discharge (142 [18%] vs 61 [8%] patients).

Ebola nurse Pauline Cafferkey 'has made full recovery'

13 November 2015



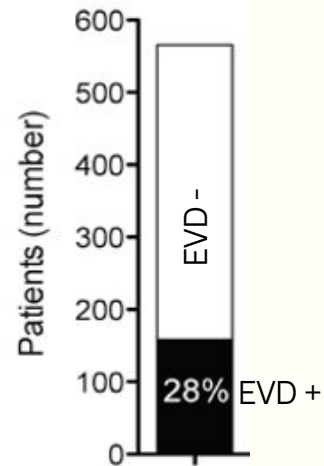
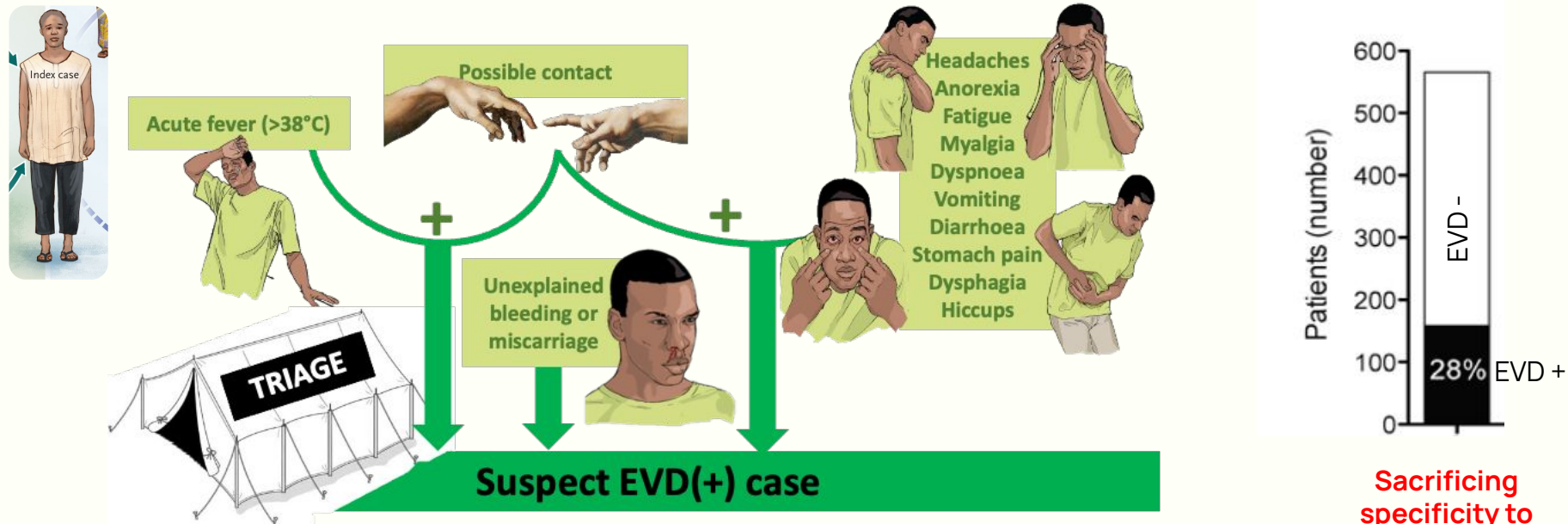
Ms Cafferkey thanked staff at the Royal Free, including senior matron Breda Athan and consultant Dr Michael Jacobs (both pictured), for the care she received

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A Scottish nurse who was readmitted to hospital after suffering complications arising from the Ebola infection has made a "full recovery", doctors say.

Ebola Virus disease (EVD)



Sacrificing specificity to prioritize sensitivity

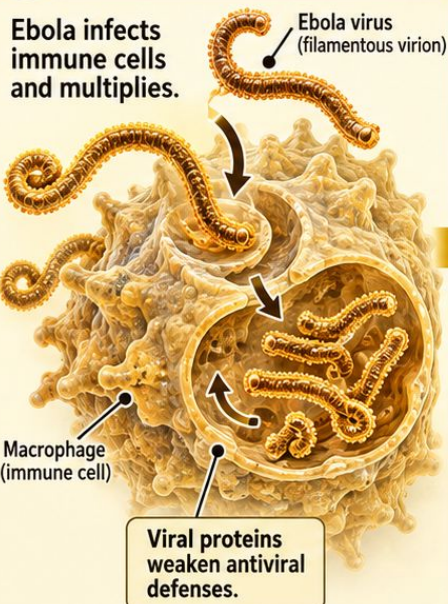
The disease

Pathophysiology

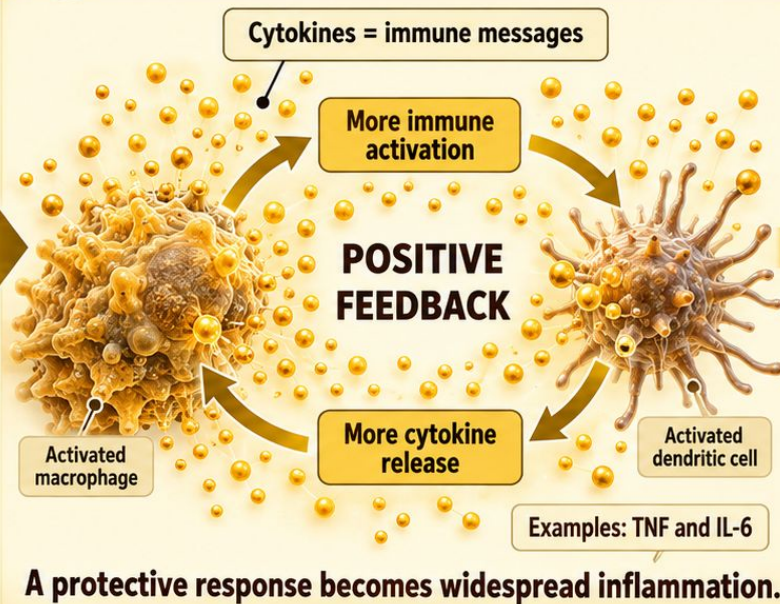
Cytokine storm

An excessive
dysregulated surge of
inflammation

1 VIRAL TRIGGER



2 RUNAWAY SIGNALING

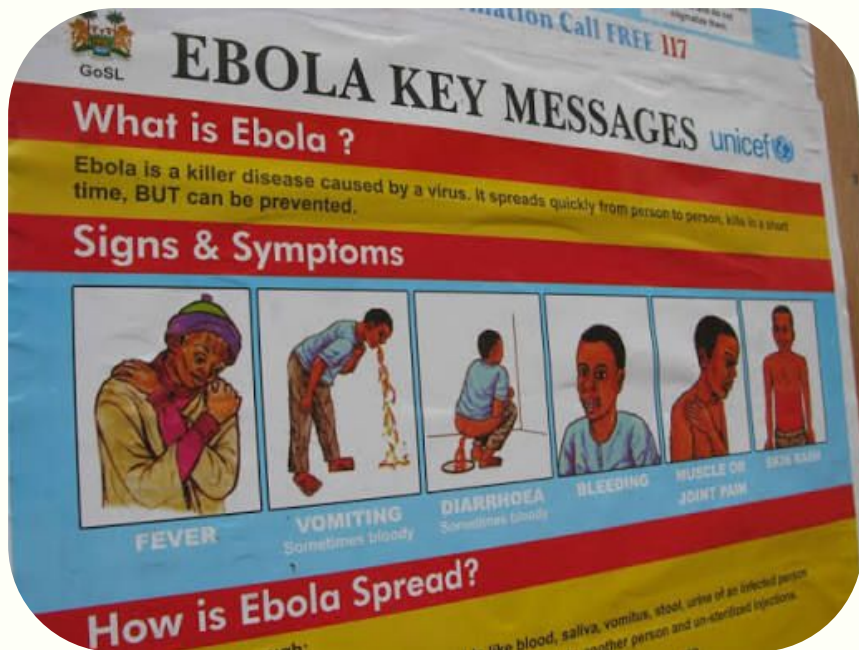


3 SYSTEM DAMAGE



Syndromic surveillance

During the outbreak



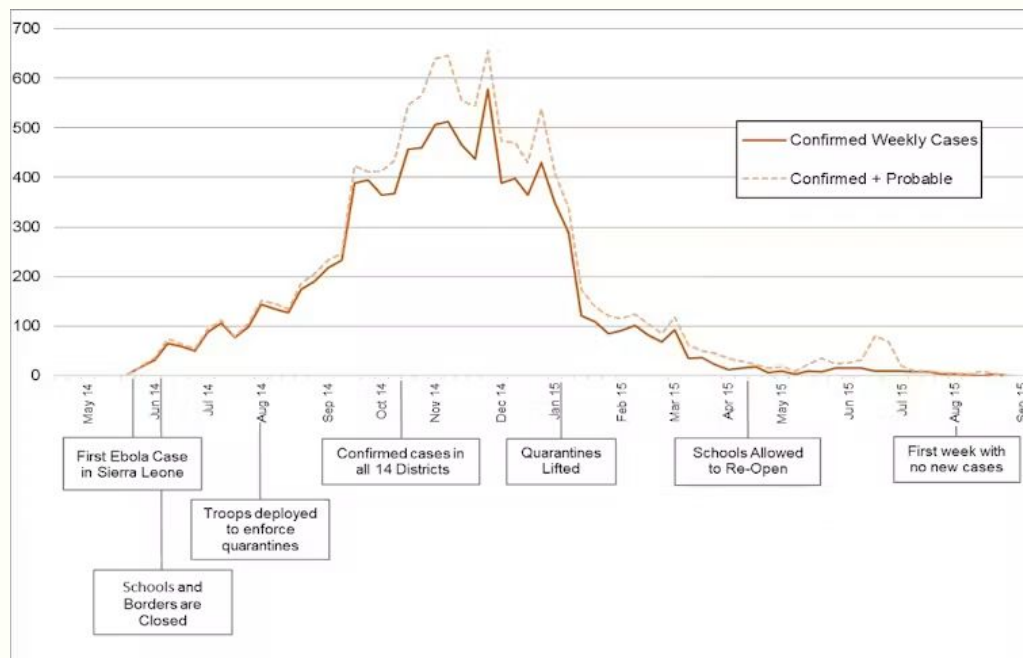
Outside of the outbreak

(community surveillance)

Alert if there is ANY one of the following

1. Fever that does not respond to treatment for the usual local causes of fever.
2. Bleeding, including bloody diarrhoea or blood in the urine.
3. Any sudden death

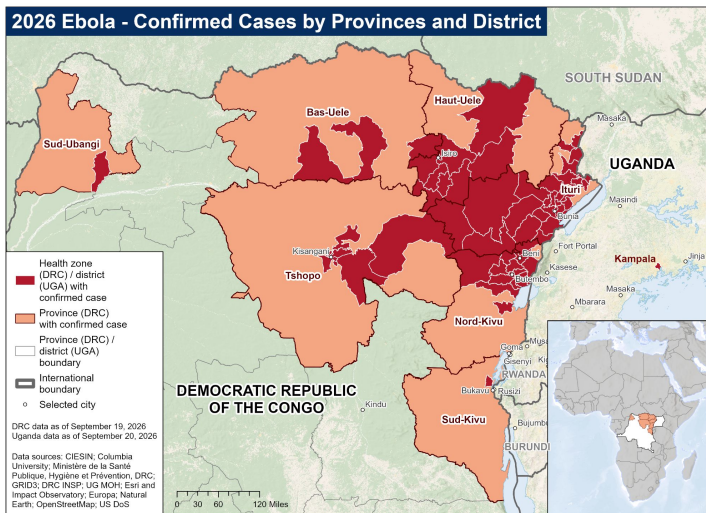
Syndromic surveillance



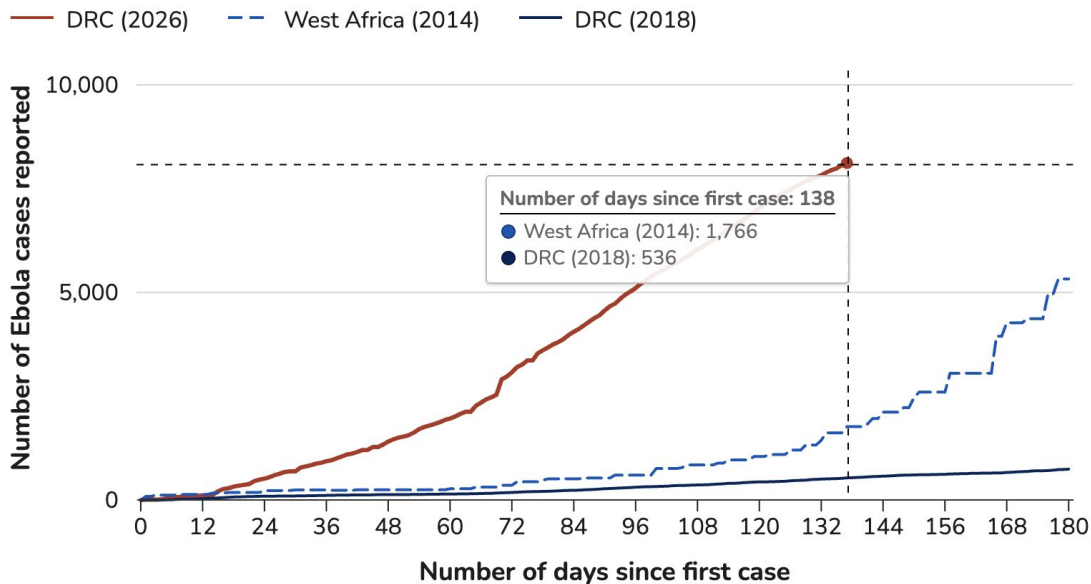
The outbreak

CONTEXT

And now in DRC...?



Cases in the first six months of Ebola outbreaks, 2014–2026

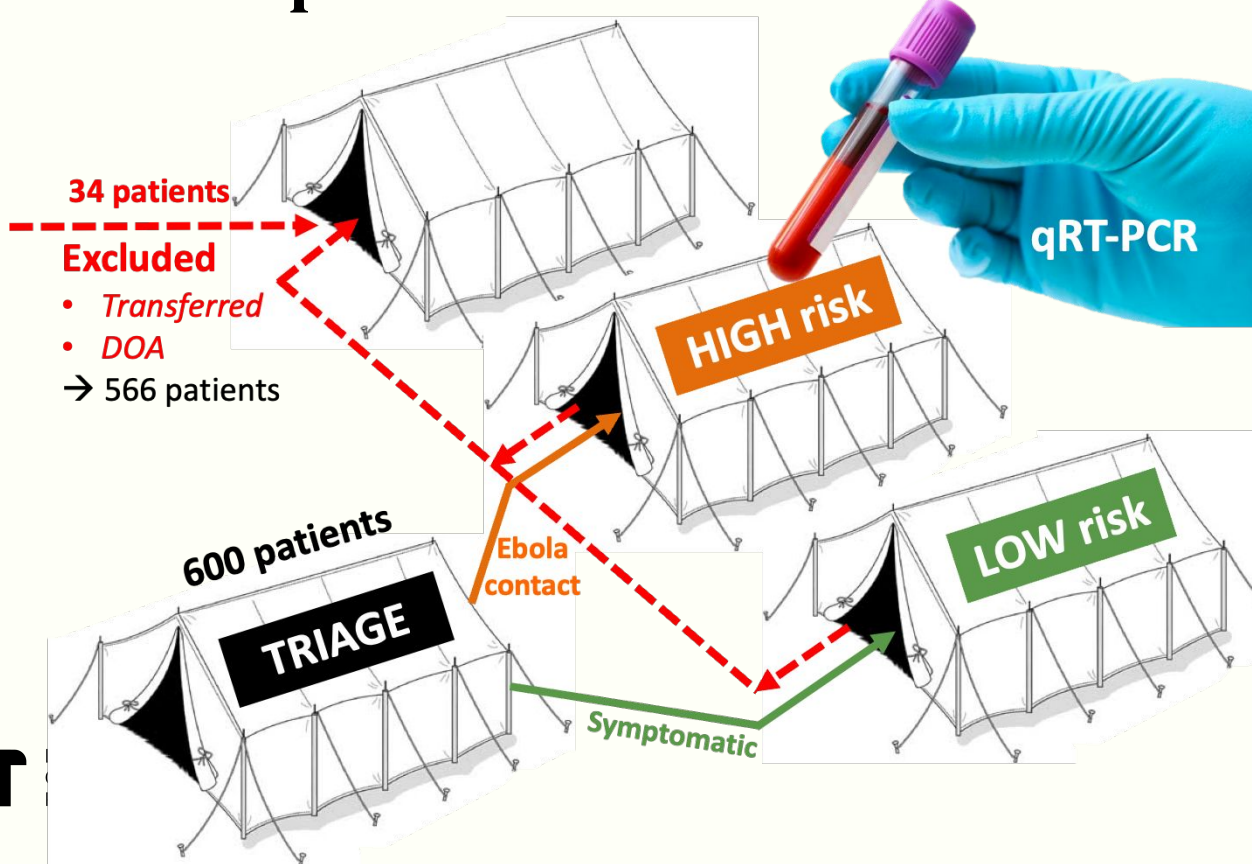


*Cases shown based on date reported. Data Sources: DRC INSP, WHO. Data for DRC (2026) are as of 2026-09-28.

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Triage: an example from Sierra leone



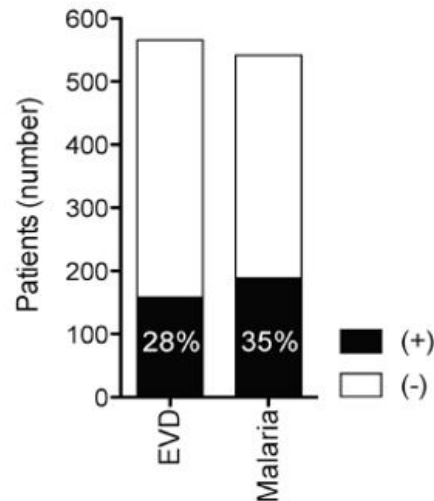
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Triage: an example from Sierra Leone

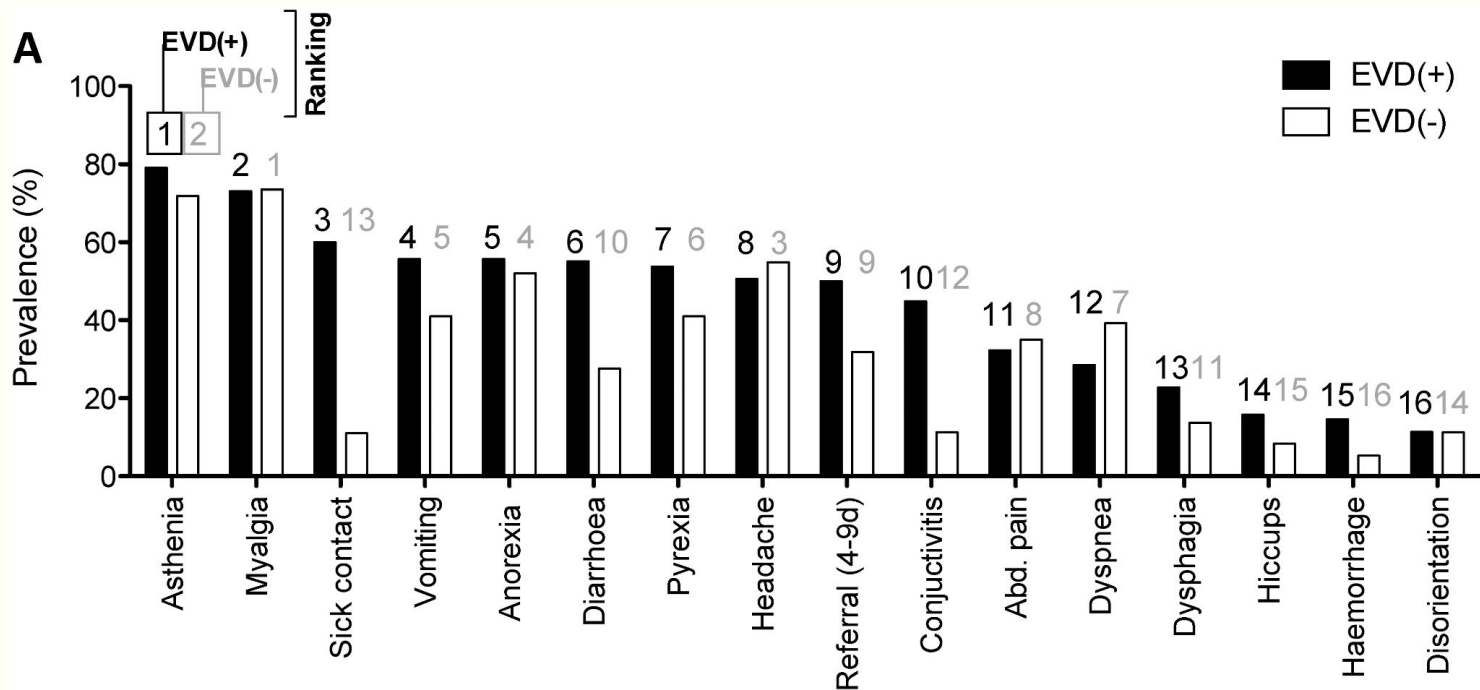
Predicting Ebola infection: A malaria-sensitive triage score for Ebola virus disease

Mary-Anne Hartley , Alyssa Young , Anh-Minh Tran , Harry Henry Okoni-Williams, Mohamed Suma, Brooke Mancuso, Ahmed Al-Dikhari, Mohamed Faouzi

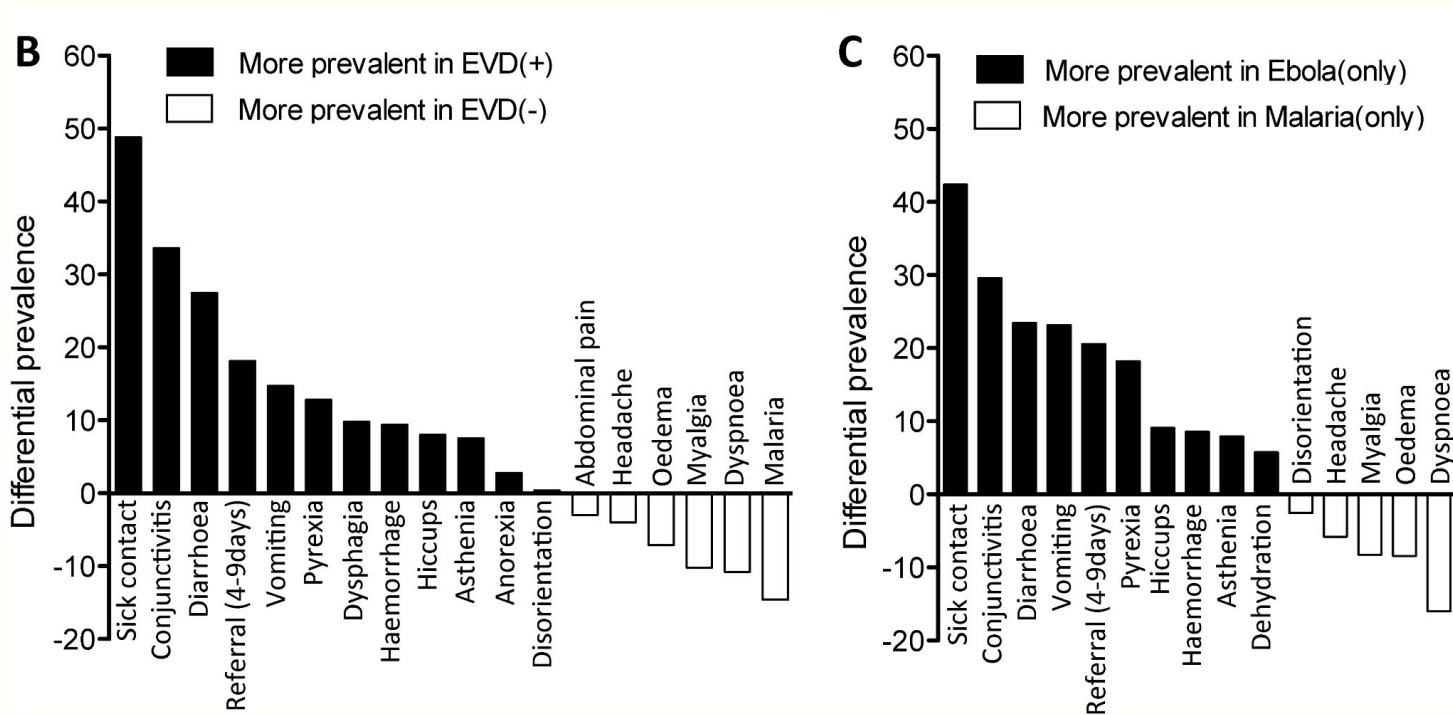
Published: February 23, 2017 • <https://doi.org/10.1371/journal.pntd.0005356>



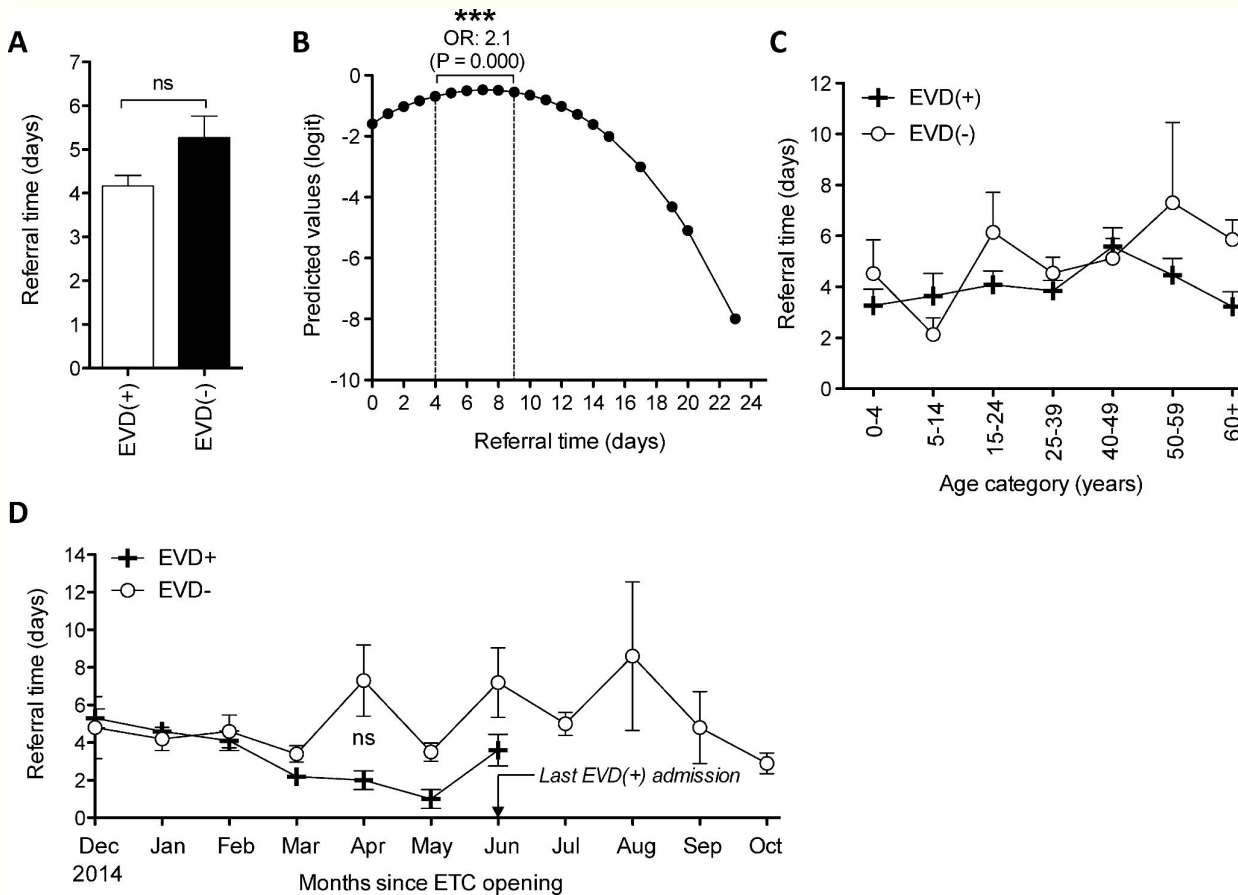
Triage



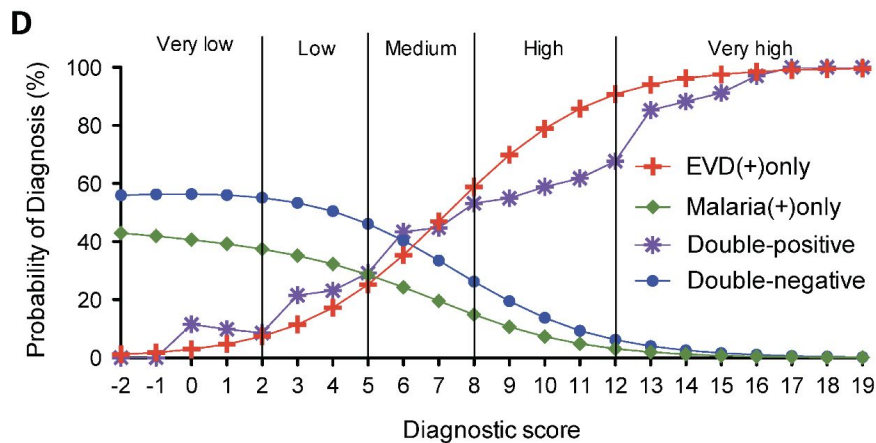
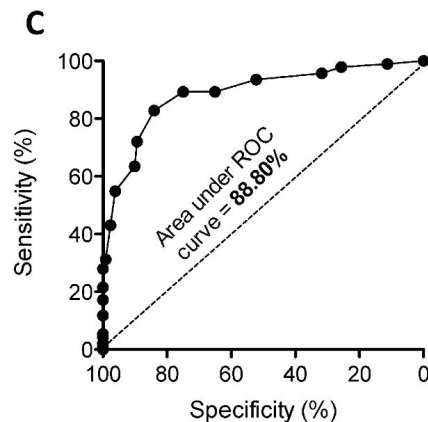
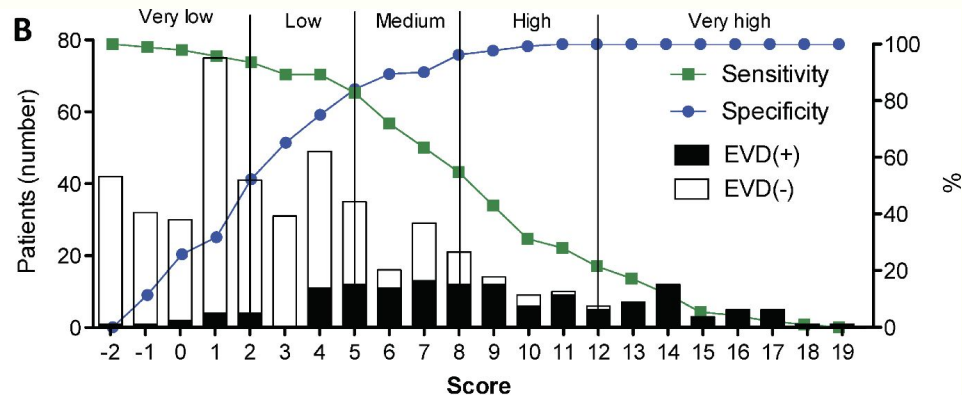
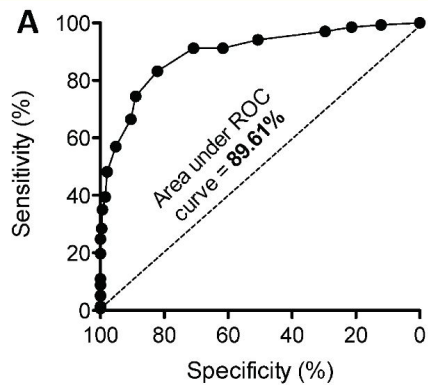
Triage



Triage

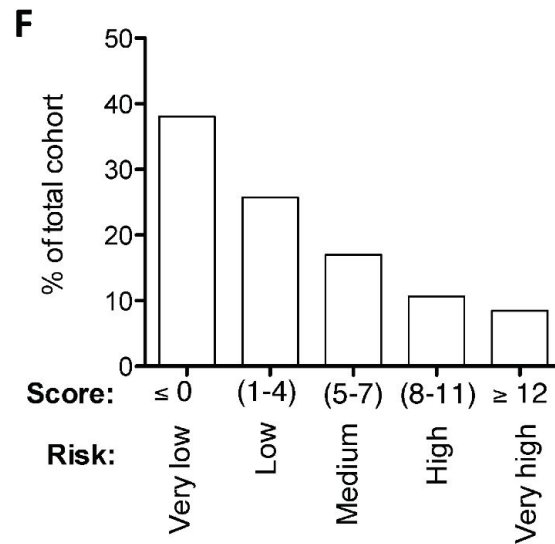
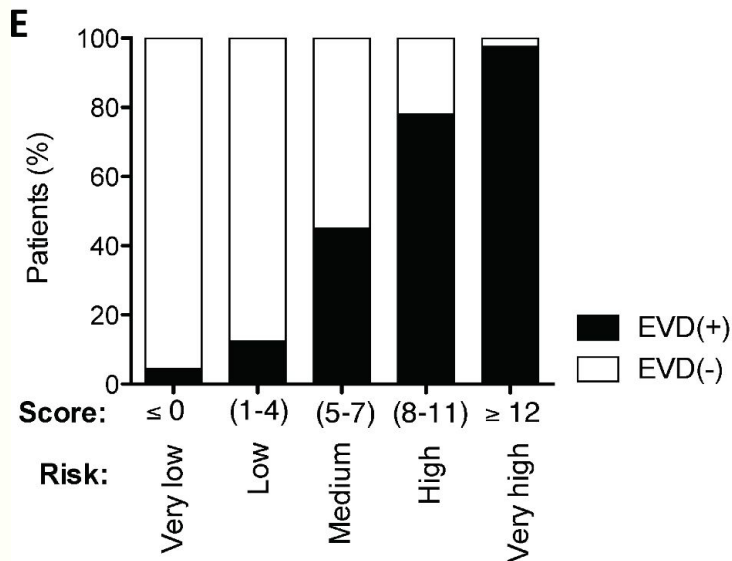


Triage

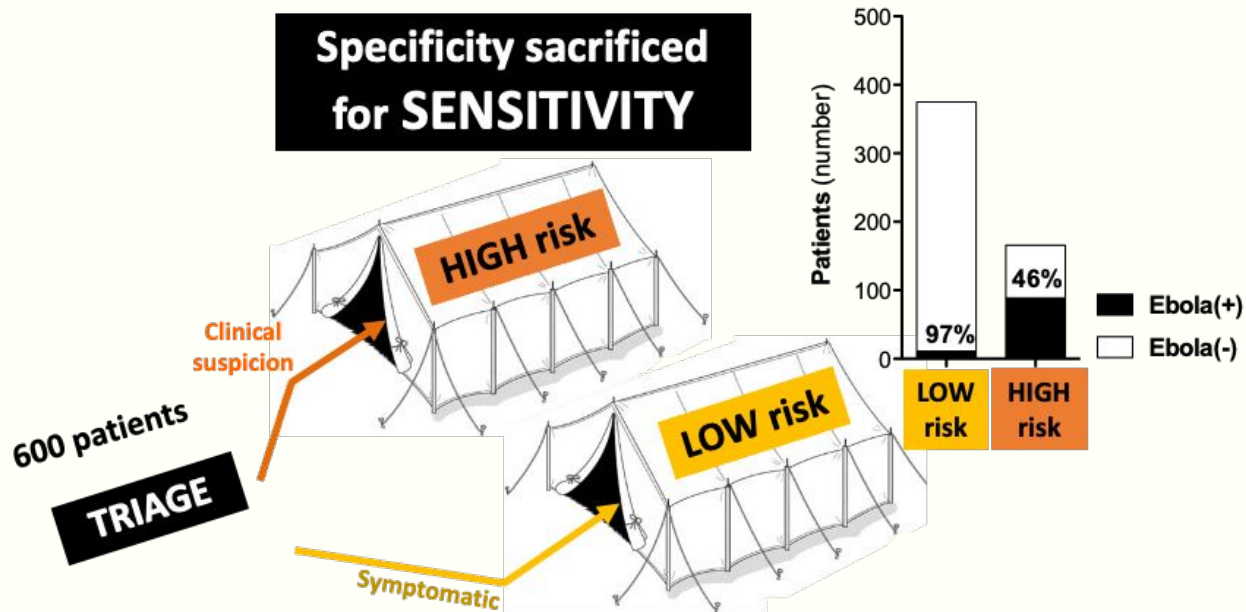
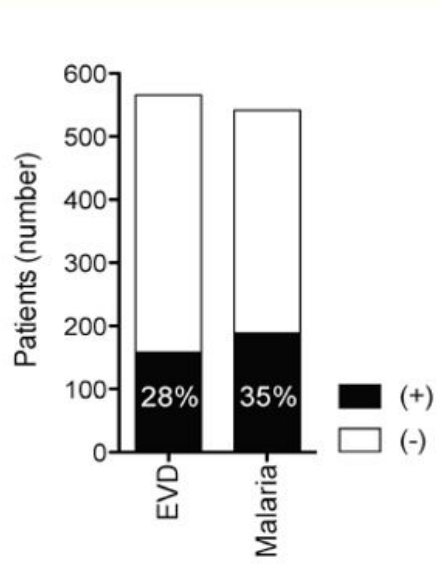


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Triage



The result

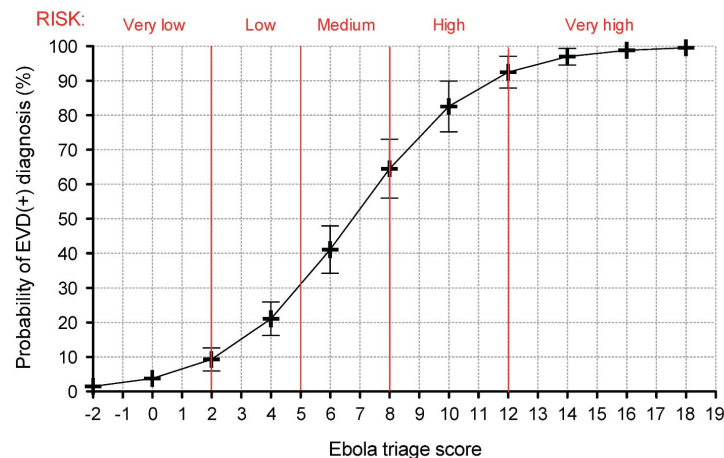


Simple solution

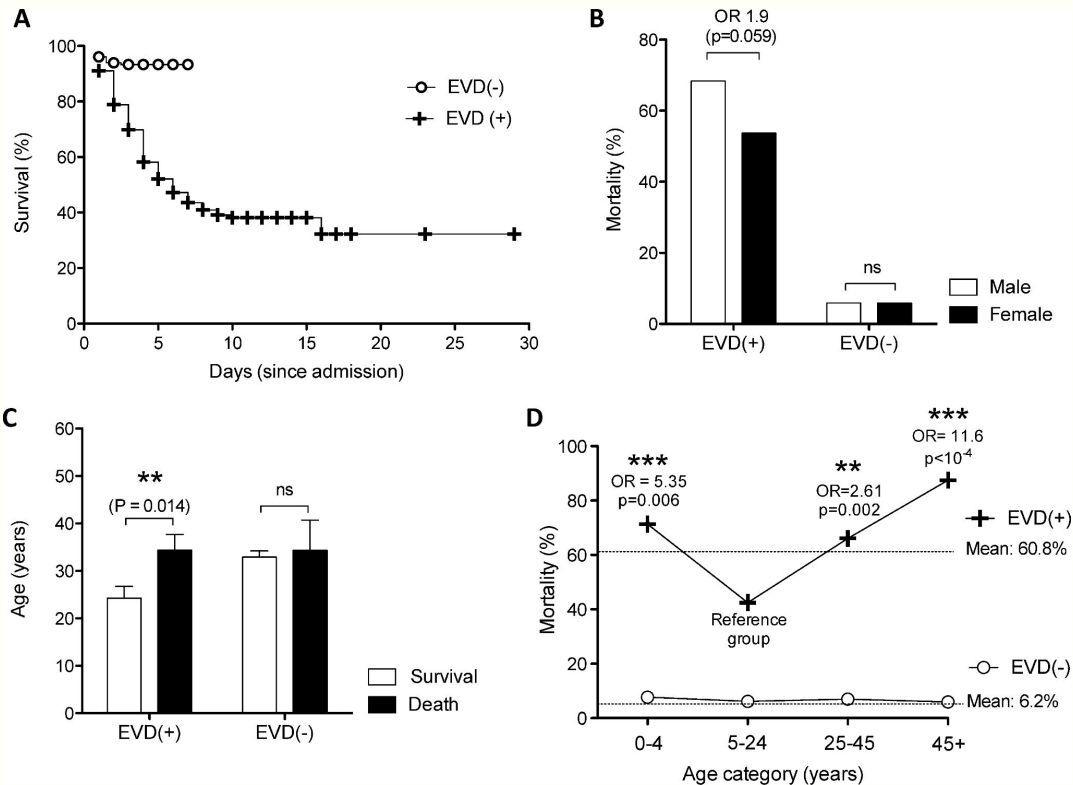
Ebola Infection Risk Score

			Calculation
1	Ebola contact history		+ 6
2	Days since 1 st symptom = 4-9 days		+ 3
3	Symptoms at triage	Conjunctivitis	+ 4
		Diarrhoea	+ 3
		Dysphagia	+ 2
		Haemorrhage	+ 2
		Temperature >38°C	+ 1
		Myalgia	- 2
TOTAL:			

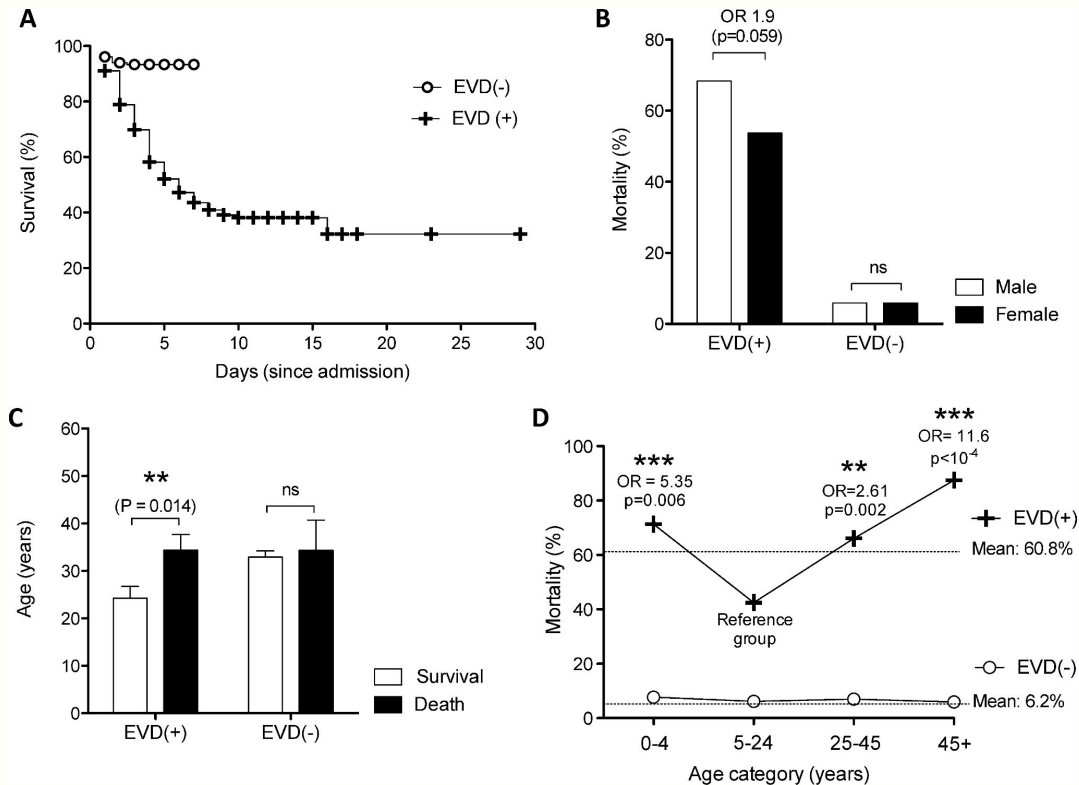
Risk of Ebola infection <i>89% accuracy</i>	Very low (3%)	Low (15%)	Medium (41%)	High (78%)	Very high (98%)
Score	≤ 1	[2-4]	[5-7]	[8-11]	≥ 12



Prioritization



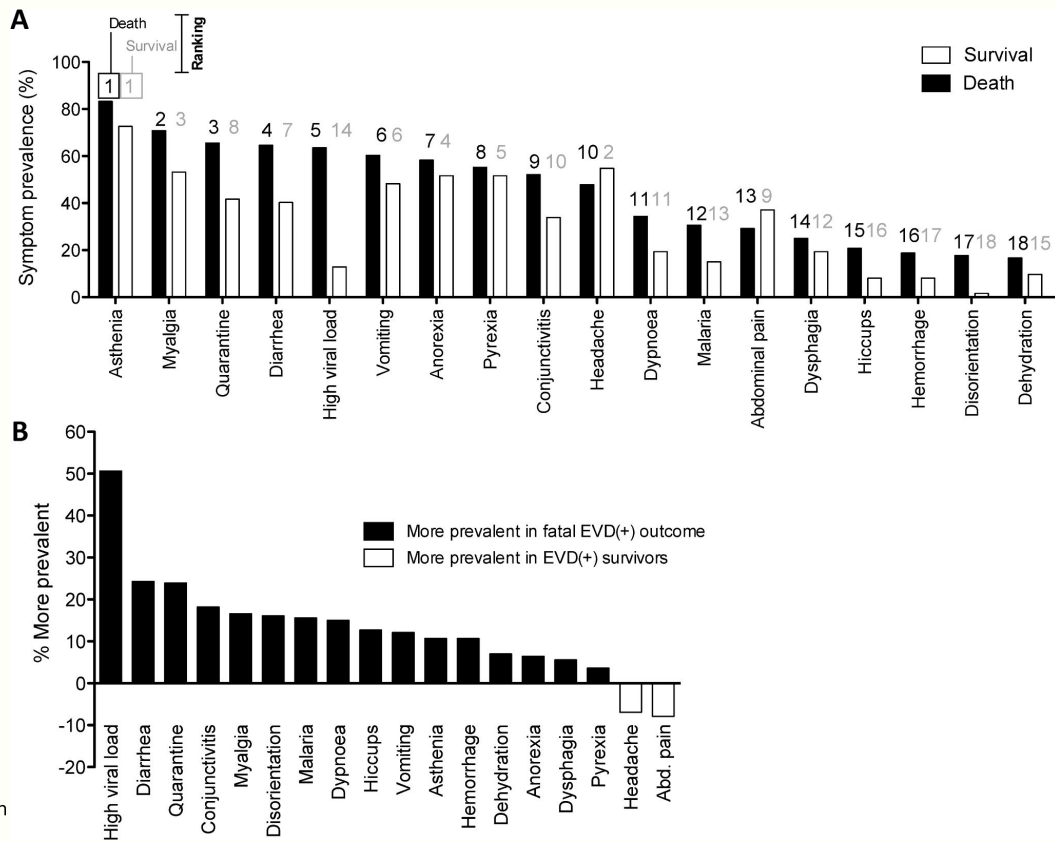
Prioritization



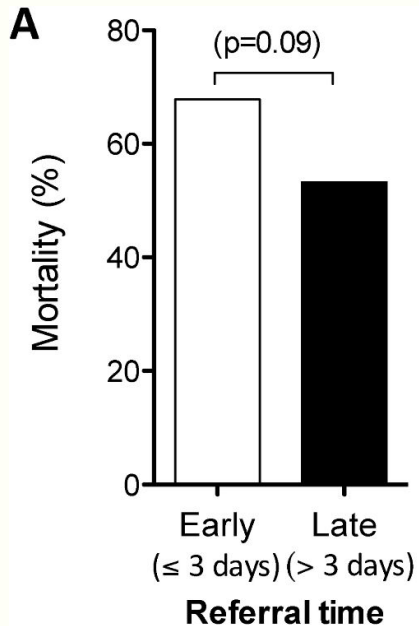
Prioritization



Prioritization

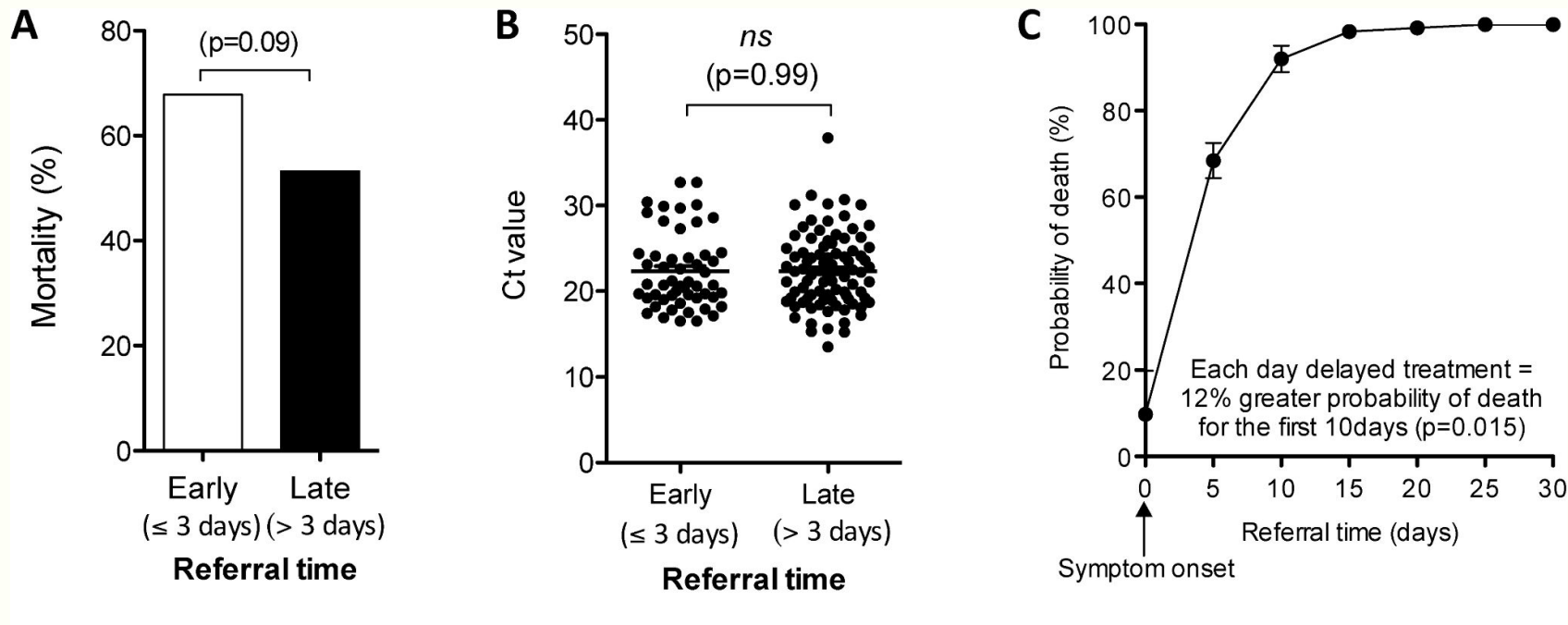


Prioritization



Ah so...coming to the Ebola Treatment facility is associated with death??
I think I'll stay at home then...

Prioritization



The outbreak

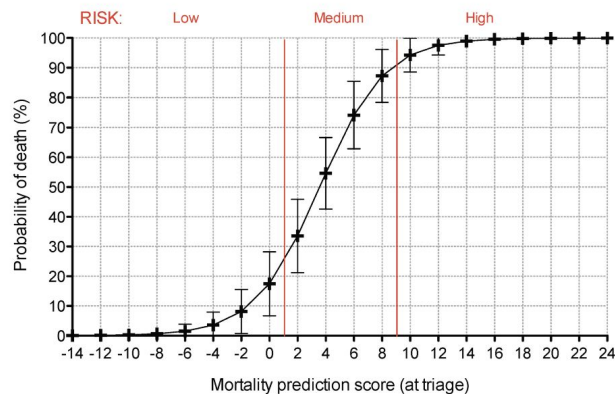
CONTEXT

A Ebola Severity Score at Triage

Ebola severity score at triage				Calculation
1	Age (years)	< 5	+ 4	
		5-25	+ 0	
		25-45	+ 4	
		> 45	+ 9	
2	Symptoms at triage	Muscle pain	+ 3	
		Disorientation	+ 7	
3	Referral time (Days since 1 st symptom)	If Ct value >20	SUBTRACT days	
		If Ct value ≤ 20	ADD days	
TOTAL:				

Risk of death 84% accuracy	Low (7%)	Medium (66%)	High (99%)
Score	≤ 1	[2-9]	≥ 10

Note: External Validation required before clinical use

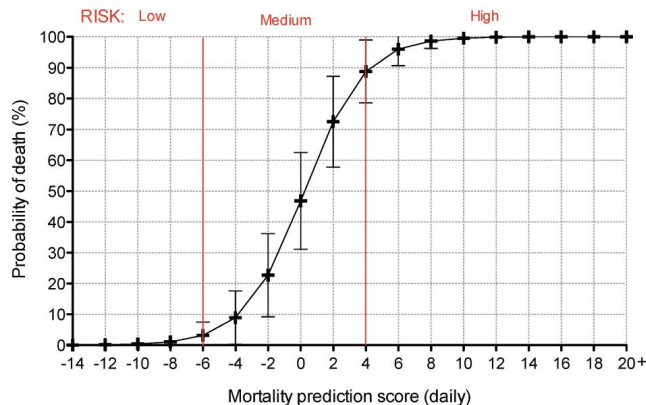


B Ebola Severity Score After Triage (daily)

Elderly Severity Score After: Widge (4444)				Calculation
1	Age (years)	< 5	+ 4	
		5-25	+ 0	
		> 25	+ 4	
2	Daily symptoms	Disorientation	+ 10	
		Haemorrhage	+ 6	
3	Days spent in ETC	SUBTRACT days		
TOTAL:				

Risk of death 93% accuracy	Low (<1%)	Medium (39%)	High (98%)
Score	≤ -6	[-5 to 3]	(≥ 4)

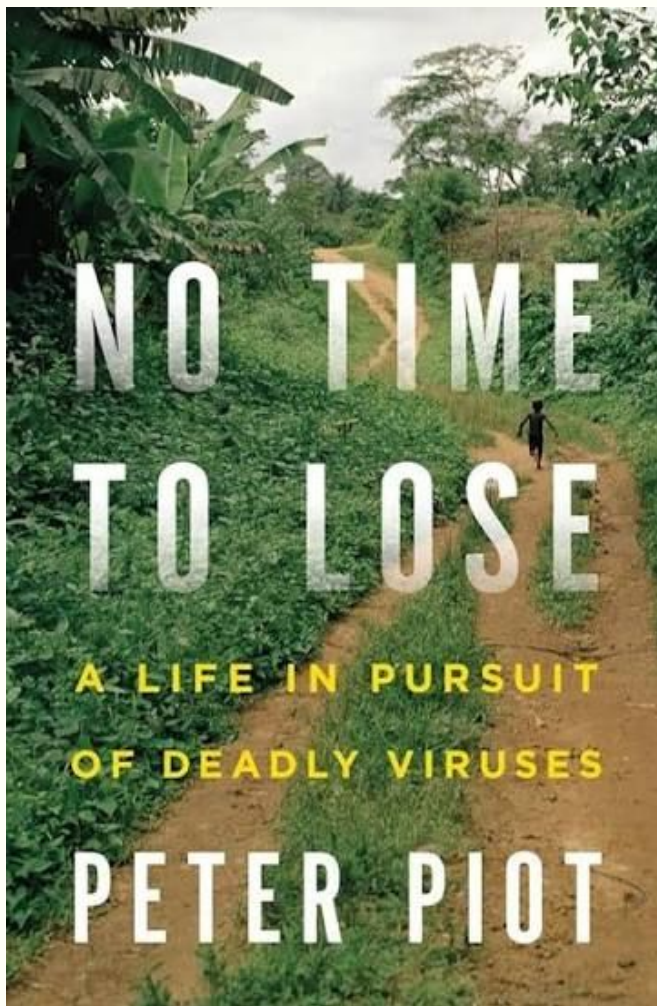
Note: External Validation required before clinical use



Ebola discovery

Must read

CONTEXT



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Listening to the World: *Epidemic Intelligence* at Scale

Lecturer: *Fabrice Nemo*

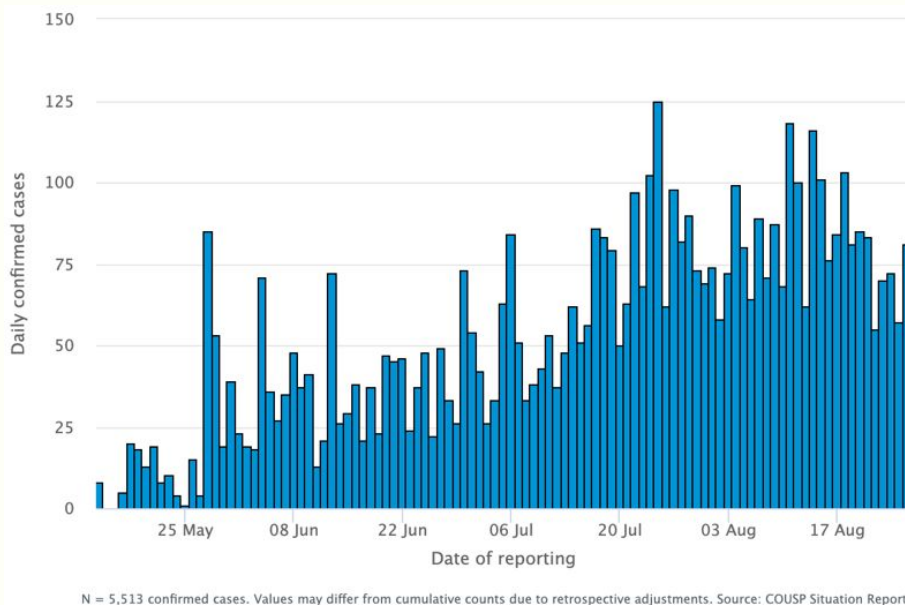
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Outline

- 01 Hook: The monitoring of the 2026 Ebola Epidemic
- 02 Epidemic Intelligence is a Pipeline
- 03 Gather Data
- 04 Analyze Data
- 05 Engineering Failure: Google Flu Trends

Hook: The monitoring of the 2026 Ebola Epidemic

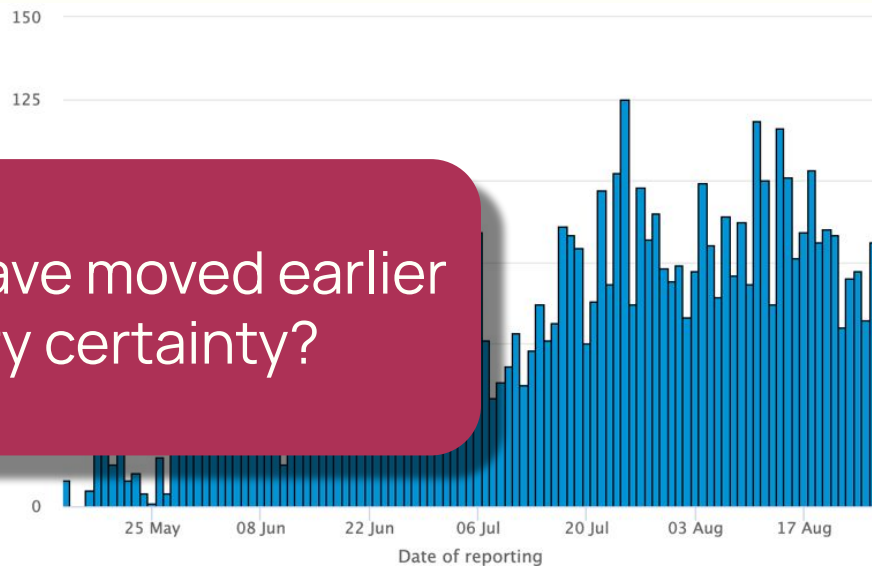


Hook: The monitoring of the 2026 Ebola Epidemic

First case

Jan 26/Feb 26?

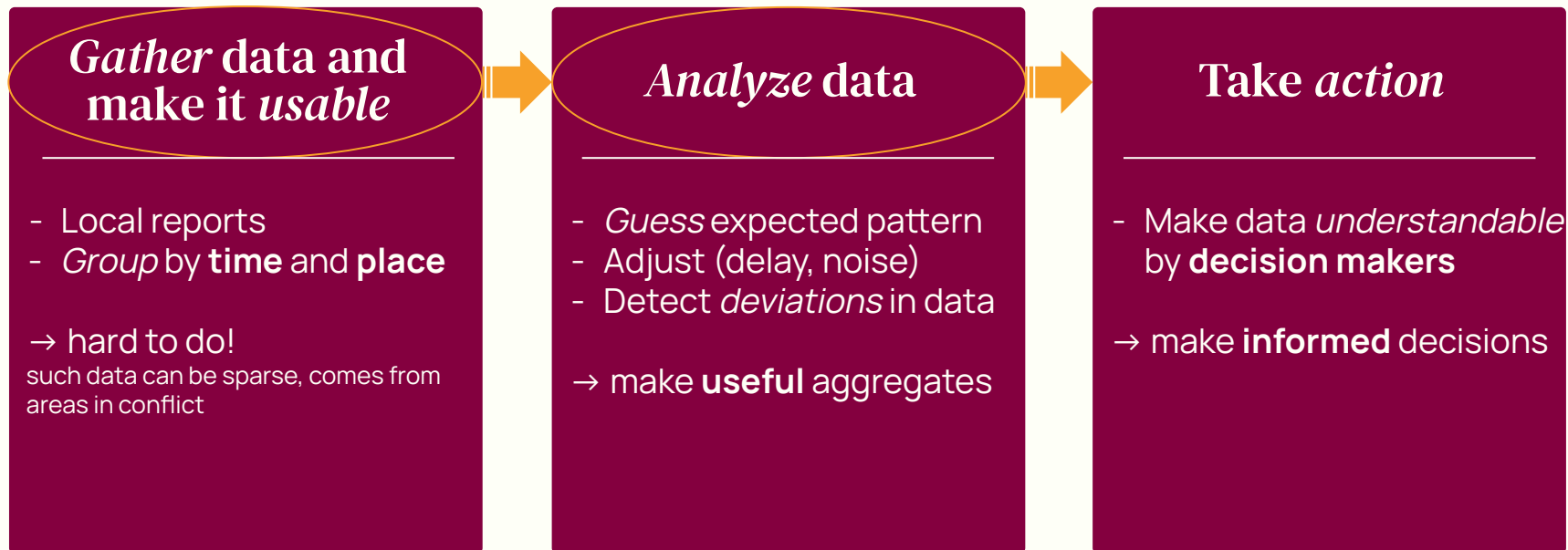
What *data* could have moved earlier
than laboratory certainty?



N = 5,513 confirmed cases. Values may differ from cumulative counts due to retrospective adjustments. Source: COUSP Situation Report.

What can we do, as *Data Scientists*, about it?

Epidemic Intelligence is a *Pipeline*



You don't see the epidemic, just biased *signals*



Define the *observations*

Press article: “5 flu-related deaths in the retirement home Domaine de la Gracieuse”

Web searches: 100 Google searches for flu-like symptoms

Measurements: 10,000 influenza RNA copies measured in wastewater

Emergency Calls: 20 calls mentioning flu-like symptoms

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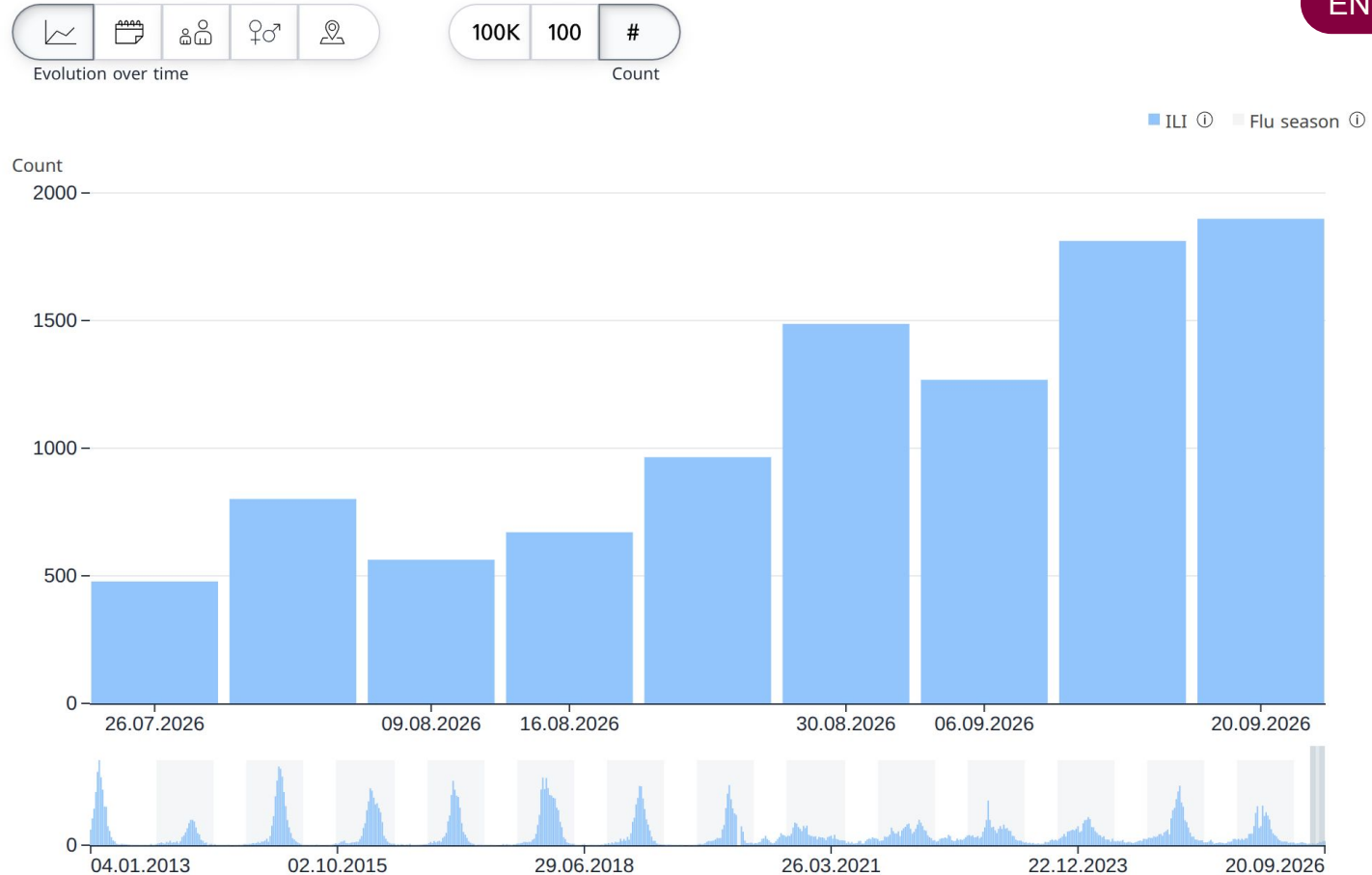
Event Time	Report Time	Location	Flu specificity
January 15	January 17	Domaine de la Gracieuse, Lonay VD	High
January 16	January 17	Lausanne Greater Area	Weak
January 14	January 21	Sewers of Ecublens VD	High
January 14	January 18	Vaud	Middle

Define the *observations*

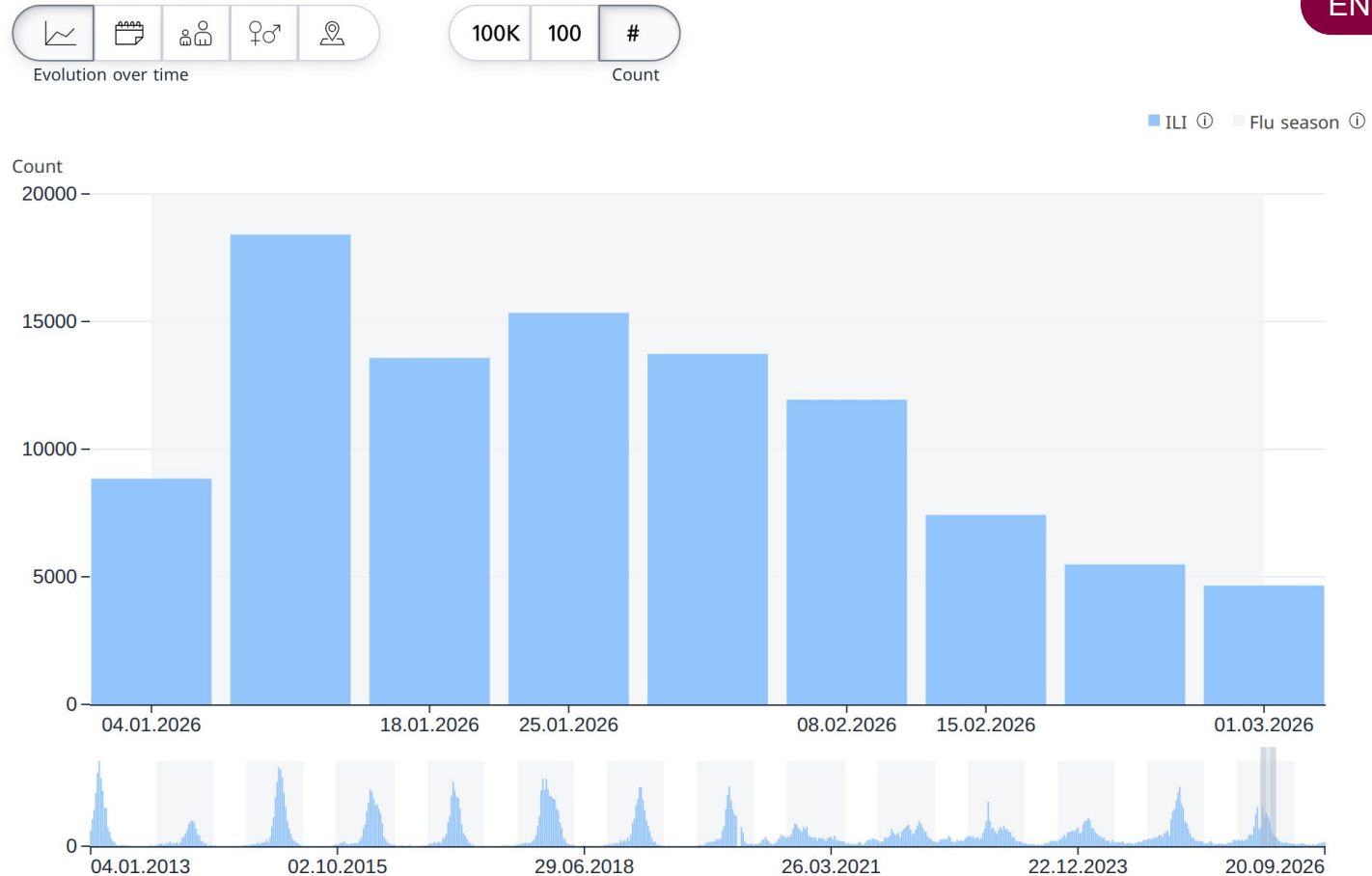
Another proxy: physician visits



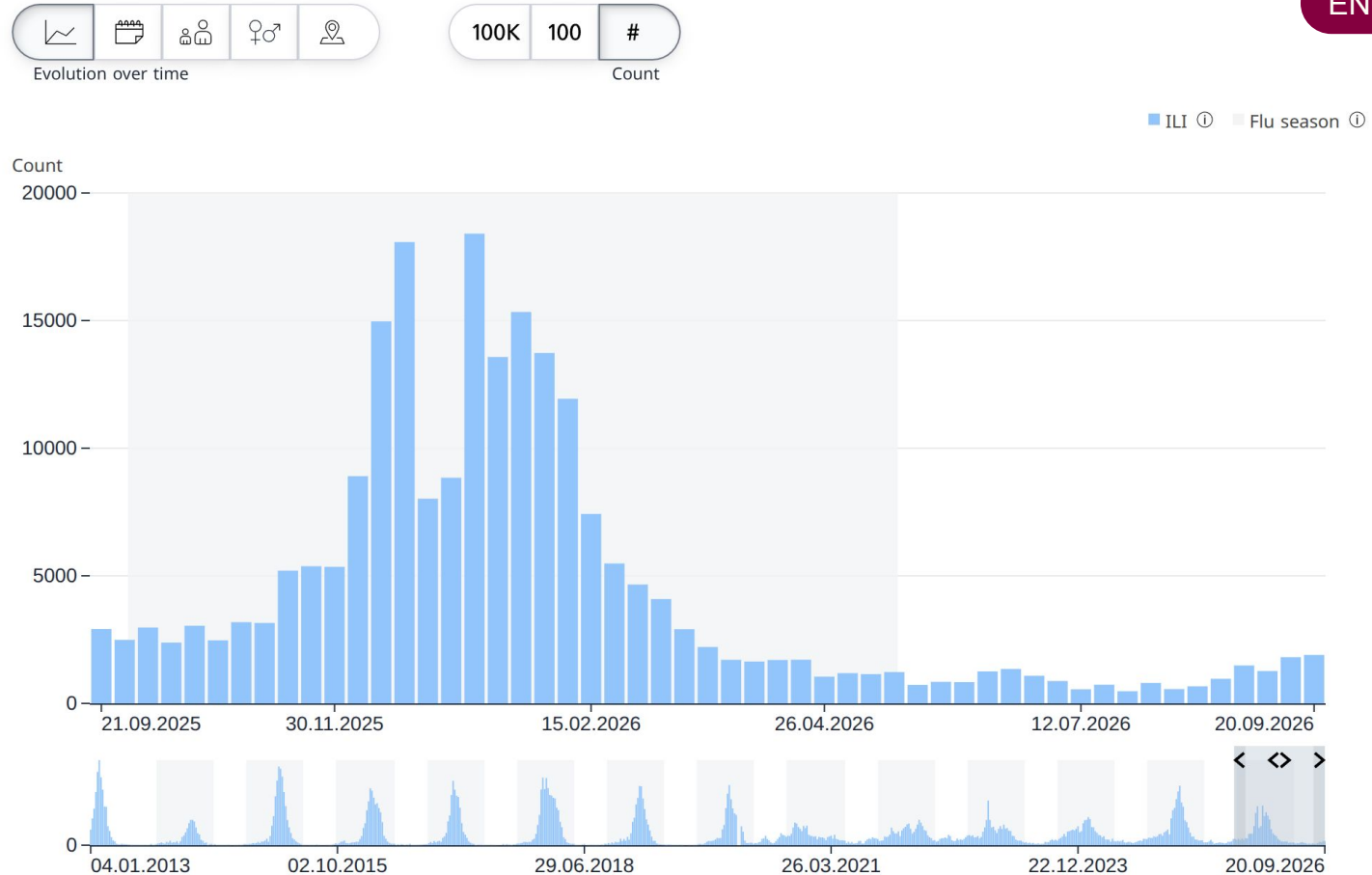
The chart shows the temporal development of the weekly number of doctor consultations for influenza-like illness (ILI), extrapolated for Switzerland.



The chart shows the temporal development of the weekly number of doctor consultations for influenza-like illness (ILI), extrapolated for Switzerland.



The chart shows the temporal development of the weekly number of doctor consultations for influenza-like illness (ILI), extrapolated for Switzerland.



What did we *expect*?

Action may be needed when we deviate from the **baseline**

→ What is the **baseline**?

What did we *expect*?

Action may be needed when we deviate from the **baseline**

→ What is the **baseline**?

Baseline → expected behavior (if nothing unusual is happening)

Deviation = Observation - Baseline

→ Need to **detect** whether we deviate from the baseline and *by how much*

→ **Investigate**, if the deviation is sufficiently large

A First *Detector*

```
# visits is a Pandas Series object, visits[t]: number of visits on week
t

window = 4
past = visits.shift(1) # exclude the current week

expected = past.rolling(window).mean()
scale     = past.rolling(window).std()

# compute the z_score for all weeks
z_score = (visits - expected) / scale
alert = z_score > 2
# btw, why 4 and 2?
```

When to give the *alert*?

1. First case

Average over the past month of **700** visits/week, std at **200**, current week at **1101**

→ $z_score = 2.005 > 2$

2. Second case

Average over the past month of **700** visits/week, std at **200**, current week at **800**

→ $z_score = 0.5 < 2$

3. Third case

Average over the past month of **700** visits/week, std at **200**, current week at **1300**

→ $z_score = 3 > 2$

When to give the

1. First case

Average over the past month

→ $z_score = 2.005 > 2$

2. Second case

Average over the past month

→ $z_score = 0.5 < 2$

3. Third case

Average over the past month

→ $z_score = 3 > 2$

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z_score
at 2.005



z_score
at 0.5



z_score
at 3



ek at 1101

ek at 800

ek at 1300

Beyond Z-score

1. EWMA

→ Detect **gradual increases**, care more about recent data than ancient data

$$s_t = \alpha z_t + (1 - \alpha) s_{t-1}$$

2. CUSUM

→ **Accumulate** persistent positive standardized deviations, small deviations add up

$$C_t = \max(0, C_{t-1} + z_t - k)$$

Interpreting EWMA and CUSUM

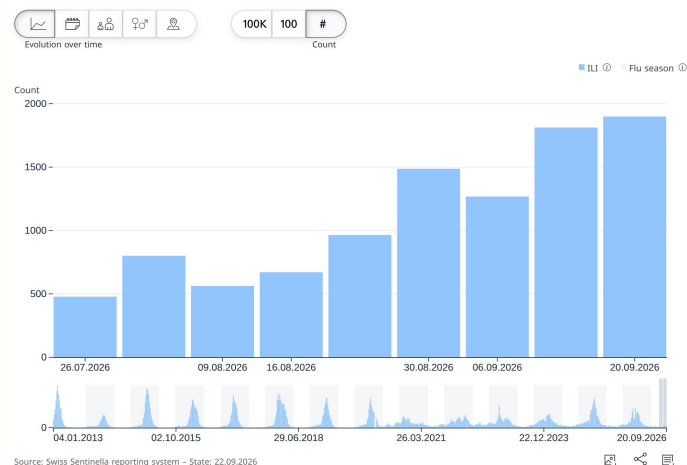
EWMA (here $\alpha = 0.3$)

→ Smooths standardized deviations

CUSUM (here $k = 0.5$)

→ Accumulates evidence above an allowance k

The chart shows the temporal development of the weekly number of doctor consultations for influenza-like illness (ILI), extrapolated for Switzerland.



Source: Swiss Sentinel reporting system - State: 22.09.2026

Model	26.07	02.08	09.08	16.08	23.08	30.08	06.09	13.09	20.09
EWMA score	-0.32	0.15	-0.04	0.06	0.77	2.38	3.04	4.67	6.00
CUSUM score	0.00	0.74	0.00	0.00	1.91	7.56	11.64	19.62	28.22

The Present is *Incomplete*

- Unavoidable report delays
- Data about the present is missing

→ **Was the available data on January 20 complete?**

Event Time	Report Time	Location	Flu specificity
January 15	January 17	Domaine de la Gracieuse, Lonay VD	High
January 16	January 17	Lausanne Greater Area	Weak
January 14	January 21	Sewers of Ecublens VD	High
January 14	January 18	Vaud	Middle to high

Text as a signal

"Several schools in Lausanne report unusually high absenteeism due to fever and cough."
January 19

"Vaud pharmacy chains report a sudden increase in sales of cough medicine."
January 16

"Emergency departments have longer waiting times for patients with respiratory symptoms."
January 17

"damn everyone in Vortex has the flu this week"
Tweet on January 18

→ Use **NLP techniques** for spotting *relevant* qualitative information

→ Filter, detect, interpret

Text as a signal - Relevance Classification Techniques

"Several schools in Lausanne report unusually high absenteeism due to fever and cough."
January 19

"Vaud pharmacy chains report a sudden increase in sales of cough medicine."
January 16

"Emergency departments have longer waiting times for patients with respiratory symptoms."
January 17

"damn everyone in Vortex has the flu this week"
Tweet on January 18

- **Keyword rules**
- Embeddings + classifier
- Zero-shot LLM classification

Text as a signal - Relevance Classification Techniques

"Several schools in Lausanne report unusually high absenteeism due to fever and cough."
January 19

"Upcoming referendum on civil service in Switzerland."
January 16

"Emergency departments have longer waiting times for patients with respiratory symptoms."
January 17

"Davos is preparing for the WEF."
January 16

- Keyword rules
- **Embeddings + classifier**
- Zero-shot LLM classification

TRAINED
CLASSIFIER

YES

NO

YES

NO

Text as a signal - Relevance Classification Techniques

"Several schools in Lausanne report unusually high absenteeism due to fever and cough."
January 19

"Upcoming referendum on civil service in Switzerland."
January 16

"Emergency departments have longer waiting times for patients with respiratory symptoms."
January 17

"Davos is preparing for the WEF."
January 16

- Keyword rules
- Embeddings + classifier
- **Zero-shot LLM classification**

LLM +
Prompt

YES

NO

YES

NO

Text as a signal - NER

"Several schools in Lausanne report unusually high absenteeism due to fever and cough."
January 19

"Vaud pharmacy chains report a sudden increase in sales of cough medicine."
January 16

"Emergency departments have longer waiting times for patients with respiratory symptoms."
January 17

"damn everyone in Vortex has the flu this week"
Tweet on January 18

→ **Named Entity Recognition** on each sample can yield relevant information

Location

Time

Symptoms / Disease

Text as a signal - NER

"Several schools in **Lausanne** report unusually high absenteeism due to **fever** and **cough**."

January 19

"**Vaud pharmacy** chains report a sudden increase in sales of **cough** medicine."

January 16

"**Emergency departments** have longer waiting times for patients with **respiratory symptoms**."

January 17

"damn everyone in **Vortex** has the **flu this week**"

Tweet on January 18

→ **Named Entity Recognition** on each sample can yield relevant information

Location

Time

Symptoms / Disease

+ others...

Text as a signal - Source Quality Varies

"Several schools in Lausanne report unusually high absenteeism due to fever and cough."

Lausanne news January 19

Don't forget to **check the source**, some are more reliable than others

"Vaud pharmacy chains report a sudden increase in sales of cough medicine."

Regional news January 16

Treat source quality as metadata

"Emergency departments have longer waiting times for patients with respiratory symptoms."

Swiss news January 17

"damn everyone in Vortex has the flu this week"

Tweet on January 18

Corroboration before alarm

corroboration

[Entry](#) [Discussion](#) [Citations](#)

English [\[edit \]](#)

Etymology [\[edit \]](#)

From [Middle English](#) *corroboracioun*, borrowed from [Late Latin](#) *corrōborātiō* ("strengthening").

Pronunciation [\[edit \]](#)

- Rhymes: -eɪʃən

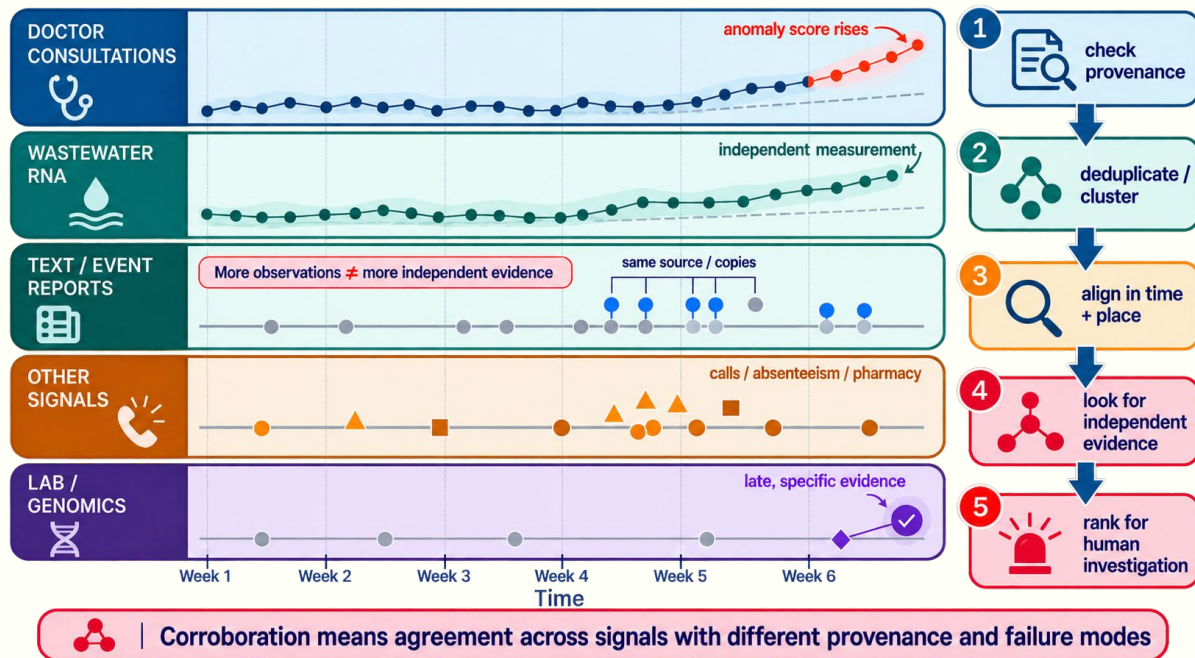
Noun [\[edit \]](#)

corroboration (*countable and uncountable*, plural **corroborations**)

1. The act of [corroborating](#), [strengthening](#), or [confirming](#); addition of strength; [confirmation](#).

Source: Wiktionary, "corroboration", CC BY-SA 4.0.

Corroboration before alarm



Google Flu Trends

Google Flu Trends

WIKIPEDIA

25 years of the free encyclopedia

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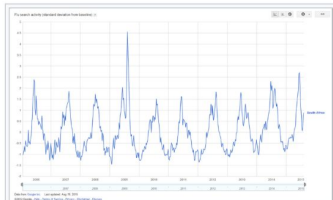
View history

From Wikipedia, the free encyclopedia

Google Flu Trends (GFT) was a [web service](#) operated by [Google](#). It provided estimates of [influenza](#) activity for more than 29 countries. By aggregating [Google Search](#) queries, it attempted to make accurate predictions about flu activity. This project was first launched in 2008 by Google.org to help predict outbreaks of flu.^[1]

Google Flu Trends stopped publishing current estimates on 9 August 2015. Historical estimates are still available for download, and current data are offered for declared research purposes.^[2]

Google Flu Trends data, South Africa



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History

[\[edit \]](#)

The idea behind Google Flu Trends was that, by monitoring millions of users' health tracking behaviors online, the large number of Google search queries gathered can be analyzed to reveal if there is the presence of flu-like illness in a population. Google Flu Trends compared these findings to a historic baseline level of influenza activity for its corresponding region and then reports the activity level as either minimal, low, moderate, high, or intense. These estimates have been generally consistent with conventional surveillance data collected by health agencies, both nationally and regionally.

[Roni Zeiger](#) helped develop Google Flu Trends.^[3]

Source: Wikipedia, "Google Flu Trends", CC BY-SA 4.0.

Google Flu Trends

Gather **easy-to-access** data that is arguably **correlated** with flu outbreaks

→ Google queries **related to flu symptoms**

→ Metadata (**time, location**)

→ Make **trends** for geographic regions over time

[Wikipedia - Google Flu Trends](#)

Google Flu Trends

Impact [\[edit\]](#)

An initial motivation for GFT was that being able to identify disease activity early and respond quickly could reduce the impact of seasonal and pandemic influenza. One report was that Google Flu Trends was able to predict regional outbreaks of flu up to 10 days before they were reported by the CDC (Centers for Disease Control and Prevention).^[11]

In the 2009 flu pandemic Google Flu Trends tracked information about flu in the United States.^[12] In February 2010, the CDC identified influenza cases spiking in the mid-Atlantic region of the United States. However, Google's data of search queries about flu symptoms was able to show that same spike two weeks prior to the CDC report being released.^[citation needed]

"The earlier the warning, the earlier prevention and control measures can be put in place, and this could prevent cases of influenza," said Dr. Lyn Finelli, lead for surveillance at the influenza division of the CDC. "From 5 to 20 percent of the nation's population contract the flu each year, leading to roughly 36,000 deaths on average."^[11]

Google Flu Trends: What went *wrong*

Accuracy [\[edit\]](#)

The initial Google paper stated that the Google Flu Trends predictions were 97% accurate comparing with CDC data.^[4] However subsequent reports asserted that Google Flu Trends' predictions have been very inaccurate, especially in two high-profile cases. Google Flu Trends failed to predict the 2009 spring pandemic^[13] and over the interval 2011–2013 it consistently overestimated relative flu incidence,^[7] predicting twice as many doctors' visits over one interval in the 2012 flu season as the CDC recorded.^{[7][14]} A 2022 study published (with commentaries) in the *International Journal of Forecasting*^[15] found that Google Flu Trends was outperformed by the *recency heuristic*, an instance of so-called "naive" forecasting, where the predicted flu incidence equals the most recently observed flu incidence. For all weeks from March 18, 2007, to August 9, 2015 (the horizon for which Google Flu Trends predictions are available), the mean absolute error of Google Flu Trends was 0.38 and of the recency heuristic 0.20 (both in percentage points; linear regression with a single predictor, the most recently observed flu incidence, had a mean absolute error of also 0.20, and the benchmark of random prediction had 1.80).

Google Flu

Accuracy [\[edit\]](#)

The initial Google paper stated subsequent reports asserted that Google Flu Trends failed to predict relative flu incidence,^[7] predicted a 2022 study published (with comparison) outperformed by the recency heuristic for recently observed flu incidence (where predictions are available), the mean absolute error was 0.20 percentage points; linear regression was also 0.20, and the benchmark of

with CDC data.^[4] However, it was only in two high-profile cases. Google Flu Trends consistently overestimated flu incidence as the CDC recorded.^{[7][14]} A 2022 study published (with comparison) Google Flu Trends was outperformed by the recency heuristic for recently observed flu incidence (where predictions are available), the mean absolute error was 0.20 percentage points; linear regression was also 0.20, and the benchmark of



Google Flu Trends: What went *wrong*

- **Searching** for symptoms $\neq$ being sick
- **Media** attention can *change* search behavior
- Search behavior naturally changed over time
- **Similar symptoms** for other conditions
- **Fear** of flu leads people to search about it
- It's a proxy, it can be useful but be critical about its output

Questions?

Thank You

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